# Eduardo Wanderley de Siqueira Araújo

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## Experience

**CIn-UFPE** Recife, Brazil
*Master's researcher (Fleet UI)* *2024 – 2026*

- Built a framework that orchestrates reproducible autonomous-navigation experiments with one or more robots in simulation (ROS 2 Jazzy, Nav2, Gazebo), with a React interface and FastAPI backend
- Built AI agents on Claude (Planner, Executor, Analyst) that run full experiment campaigns from plain-language requests, with one agent per robot and isolation enforced in code
- Compared LLM multi-robot mission allocation with classical allocators: given a distance table, the LLM reached the optimum in 100% of cases, versus up to +106% cost for fixed dispatch
- Validated repeatability in simulation over 10 replays (mean RMSE 3.35 cm) and fixed simultaneous startup of 2 robots by switching the DDS layer (success from 1/5 to 10/10)

**Engeletra** Recife, Brazil
*ERP Developer (freelance)* *05/2026 – present*

- Designed and built the company ERP, now in production, with 10+ modules on Electron, React and FastAPI and a domain modeled in PostgreSQL 17, SQLAlchemy 2 and Alembic

**Empetur, Government of Pernambuco** Recife, Brazil
*Systems Analyst* *09/2025 – present*

- Built Python services that collect infrastructure metrics into management dashboards; maintain the Next.js intranet and deploy on-premises applications with Docker

**Campello & Siqueira** Recife, Brazil
*Systems Analyst / Data Engineer* *01/2024 – 04/2025*

- Led ETL pipelines and Apache Airflow DAGs with SLAs and alerts, and optimized queries on BigQuery and PostgreSQL

**Softex residency at UFPE** Recife, Brazil
*Systems Analyst (Resident)* *01/2023 – 01/2024*

- Built a dataset and computer-vision model that guides a robotic arm, plus Flask backends with REST APIs and WebSockets for real-time telemetry

## Projects

**Internal support chatbot** | *LangChain, FAISS, FastAPI, PostgreSQL, Docker, Datadog*

- Support and internal-knowledge assistant with semantic search (RAG) and monitoring.

**Educational data mining (Master's)** | *Python, Pandas, Logistic Regression, Decision Trees*

- School performance across ~5,500 Brazilian municipalities (SAEB) with interpretable models and rules for education managers.

## Education

**CIn-UFPE** 2024 – 2026
*M.Sc. in Computer Science (in progress)*

**CIn-UFPE** 2023
*Graduate certificate in Robotics and AI Applied to Software Testing*

**University of Pernambuco (UPE)** 2011 – 2021
*B.Eng. in Automation and Control Engineering*

## Publications

Mobile Robotics for Throughput Testing in a Reconfigurable Test Environment. E. Araújo and B. Miranda. SBES 2025, Brazilian Symposium on Software Engineering. DOI 10.5753/sbes.2025.11613.

## Technical Skills

**Applied AI:** AI agents, LLMs (Claude, Anthropic API), tool calling, RAG, FAISS, LangChain, supervised ML
**Backend and data:** Python, FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, BigQuery, PySpark, Airflow
**Robotics:** ROS 2 Jazzy, Nav2, SLAM Toolbox, Gazebo Harmonic, TurtleBot4, MCAP
**Infrastructure:** Docker, GitHub Actions, pytest, Linux, Datadog, GCP, AWS, React, Next.js
**Languages:** Portuguese (native), English (fluent), Spanish (intermediate), German (basic, A1)